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Predicting Physical Activity and Sedentary Behaviors After Cardiac Rehabilitation: An Observational Study

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Abstract

Background: A growing body of evidence suggests that low levels of physical activity (PA) and high levels of sedentary behavior (SB) are associated with an increased risk of adverse health outcomes in patients. Despite the objective of cardiovascular rehabilitation (CR) programs to increase PA and decrease SB levels, many patients continue to show low PA and high SB levels following their discharge from CR. The objective of the present study was to examined the levels of PA and SB following an outpatient CR program and to examine their motivational and emotional health-related correlates. PA and SB levels, along with motivational and health-related characteristics, were compared between outpatients and control participants who did not require care.

Methods: The study included 119 participants: 68 CR patients (M_{age} 57.76±10.76; 86.76% males) and 51 control participants matched on age (M_{age} 57.35±6.33 years; 45.10% males). Accelerometers were used to assess PA and SB during the first week post-discharge for outpatients and during a typical week for controls. Validated scales were employed to measure motivational (i.e., perceived capabilities, affective and instrumental attitudes, intention, approach-avoidance tendencies) and emotional health-related (i.e., anxiety, depressive symptoms, fatigue, pain intensity) factors.

Results: CR patients spent an average of 60.21 (±34.79) min of moderate-to-vigorous PA (MVPA), and 548.69 (±58.64) min of SB per day. They engaged in 18 more minutes of MVPA per day than controls (p =.038). Univariate and multivariate regressions indicated that positive affective attitudes were associated with higher MVPA (b = 10.32, R^2 = 0.07, p =.029). Additionally, males spent more time in SB than females (b = 40.54, R^2 = 0.09, p =.045). Univariate and multivariate logistic regressions showed that meeting the World Health Organization's weekly guidelines for MVPA was associated with higher perceived capabilities toward PA and more positive affective attitudes (OR = 1.17, p =.030; OR = 1.26, p <.001, respectively). Interaction tests showed no significant differences in these results between outpatients and controls.

Conclusion: These findings underscore the importance of motivational factors in PA levels following a CR program.

Introduction

Physical activity (PA), defined as “any bodily movement produced by skeletal muscles that results in energy expenditure [1]”, is an integral part of cardiac rehabilitation (CR) programs [2, 3]. Exercise-based CR has been shown to provide numerous benefits, including increased exercise capacity and quality of life, as well as decreased cardiovascular mortality and the risk of hospital readmissions [3, 4]. Conversely, sedentary behavior (SB), defined as any waking behavior characterized by an energy expenditure ≤ 1.5 Metabolic Equivalent Tasks, such as sitting and lying down [5], has been associated with several adverse health outcomes in patients. Evidence has shown that patients with coronary artery disease who adopt high levels of SB (4–8 hours per day) have a 62% higher mortality rate compared to

those who adopt in SB for less than 4 hours per day [6]. Although CR programs aim to increase PA and decrease SB levels, many patients still exhibit low PA and high SB levels after discharge from CR [7–9]. For example, only 20% of CR patients met the PA targets of 150–300 min of moderate-to-vigorous PA (MVPA) per week[10] six months after discharge [7]. Furthermore, Bakker et al. [9] showed that cardiovascular disease patients adopt higher SB levels before and up to 2 months after participating in a CR program, in comparison to healthy controls. However, while PA and SB have demonstrated independent health effects [11–13], few studies have examined these two behaviors simultaneously. Thus, measuring PA and SB levels in patients after discharge from CR and identifying their correlates is warranted.

Anchored in a dual-process perspective, it has been argued that two types of cognitive processes can influence PA behaviors [14–16]. The first type, referred to as deliberative process, is intentional, effortful, controllable, and conscious [17]. The second type, referred to as automatic process, is typically defined as unconscious, effortless, unintended, and uncontrollable [18]. Deliberative motivational factors, such as intention, perceived capability to engage in a behavior, or attitudes, are key for predicting engagement in exercise and PA and reducing SB levels [15, 19]. In the context of CR, Bierbauer et al. [8] showed that higher intention toward PA was associated with greater PA levels after discharge from CR. In the same vein, Bermudez et al. [2] found that patients with higher affective attitudes toward PA (i.e., evaluating PA as pleasant) had higher intentions toward PA and participated in more light-intensity PA after discharge from CR. In addition to these deliberative motivational factors, recent studies have shown that automatic motivational factors may further explain patients' engagement in PA. For instance, Chevance et al. [20] found that a positive automatic affective evaluation toward PA was significantly associated with higher PA levels ($\beta = .29$) after pulmonary rehabilitation. Recently, Cheval et al. [21] suggested that approach-avoidance tendencies, which refer to the automatic preparation of the organism to execute a motor pattern in the direction of a behavior [22] ,could partly explain PA engagement after a CR program.

In addition to motivational factors,, the presence of depression, anxiety, or pain, has also been associated with lower levels of PA following a CR program [7]. In the context of patients with advanced cancer, with chronic kidney disease, and with chronic obstructive pulmonary disease, perceived fatigue appeared to be a substantial barrier to PA [23–25]. However, the extent to which fatigue, PA, and SB are related in patients with cardiovascular disease after discharge from CR remains unknown. Taken together, these studies suggest that considering both deliberative and automatic motivational factors for PA, as well as emotional health-related factors, may provide a comprehensive perspective on PA and SB levels after CR. However, to the best of our knowledge, the role of automatic motivational processes in the regulation of PA and SB among patients with cardiovascular disease has been largely neglected. Finally, while previous work has shown that emotional health-related factors are associated with PA levels [7], little is known about the relationship between emotional health-related variables and SB levels.

The primary objective of this study was to assess accelerometer-based PA and SB levels during the first week following a 6-week outpatients CR program and to examine their motivational (i.e., intention, perceived capability, attitudes, and approach-avoidance tendencies) and emotional health-related (i.e.,

depressive symptoms, anxiety, fatigue, and pain intensity) correlates. Additionally, it has been suggested that the key motivational process involved in the regulation of PA and SB may differ between CR patients and a healthy control population [21, 26]. Therefore, an additional aim of this study was to compare the PA and SB levels, as well as the motivational and health-related characteristics, of patients following an outpatient CR program, with those of community-dwelling control healthy participants, who did not require care.

Methods

Study Design and Participants

The sample size was initially estimated to ensure sufficient power to detect an effect in the Improving Physical Activity (IMPACT) trial (see Cheval et al. [21]) for the IMPACT trial protocol. The IMPACT trial, carried out at Geneva University Hospitals (Switzerland), is a phase 3 single-center, placebo, triple-blind randomized controlled trial that enrolled participants from an outpatient CR program. The detailed content of the CR program is described in Supplementary Material 1. The study was approved by the Ethics Committee of Geneva Canton, Switzerland (reference number: CCER2019-02257).

The present study analyzed a dataset consisting of 68 CR patients. Furthermore, 51 additional community-dwelling control participants were recruited using snowball sampling. Community-dwelling participants were included if they were 40 years or older, matching the minimum age of outpatients, had no contraindication to PA, and were not receiving medical care. PA and SB levels were measured via accelerometer (Actigraph GT3x+; Pensacola, USA) for seven consecutive days during the week following discharge from the CR program for the outpatients and for seven consecutive days for the control participants. Motivational and emotional health-related variables were assessed before measuring PA and SB for all participants.

Measures

Physical Activity and Sedentary Behavior Outcomes

The usual level of PA, defined as the amount of time spent being physically active in a typical week during free time, was assessed using the Saltin-Grimby PA Level Scale [27]. Outpatients were asked to report their PA levels for a typical week before the onset of the health problem that necessitated the CR. Participants were categorized as (a) physically inactive, (b) engaging in some PA, (c) engaging in regular PA, or (d) engaging in regular high-intensity training.

Participants were asked to wear the accelerometer only during waking hours. Daily data were included if wearing time was ≥ 10 waking hours per day [28]. Data were included if at least 4 days met the aforementioned conditions and included one weekend day [29]. Additionally, daily PA and SB levels were self-reported using the short form of the International Physical Activity Questionnaire [30]. We used accelerometer-based daily time spent engaging in PAs of light, moderate, vigorous, and moderate-to-

vigorous intensity, as well as the number of steps, as indicators of PA levels. Finally, accelerometer-based daily time spent engaging in SB was used as an indicator of SB levels.

Motivational Factors

Perceived capability to engage in PA was assessed using the following item from the Patients-Reported Outcomes Measurement Information System (PROMIS [31]): “To what extent are you able to perform daily physical activities such as walking, climbing stairs, or moving a chair?”. Participants answered on a scale from 1 (*not at all*) to 5 (*completely*).

Intention to be physically active was assessed using the following item: “I intend to do more physical activity in the near future [21].” Participants answered on a scale from 1 (*totally disagree*) to 10 (*totally agree*).

Instrumental and affective attitudes were assessed using the following two items “Would you say physical activity is something …”. Participants answered each item on a scale from 1 (*useless*) to 10 (*useful*) for instrumental attitudes, and from 1 (*unpleasant*) to 10 (*pleasant*) for affective attitudes [32, 33].

Automatic approach-avoidance tendencies toward PA and SB were assessed using the Visual-Approach/Avoidance-by-the-Self-Task (VAAST [34]). The VAAST protocol is explained in more detail in Supplementary Material 2.

Emotional Health-Related Factors

Anxiety, depression, fatigue, and pain intensity were assessed using PROMIS items. Anxiety was assessed using the following two items: “In the past 7 days, I felt scared” and “In the last 7 days, I found it hard to focus on anything other than my anxiety." Participants answered each item on a scale from 1 (*never*) to 10 (*always*). Items were averaged (Pearson correlation = 0.61).

Depression was assessed using the following four items: “In the past 7 days, I felt..." "useless," "powerless," "desperate," and "depressed." Participants answered each item on a scale from 1 (*never*) to 10 (*always*). The items were averaged (Cronbach's α = 0.81).

Fatigue was assessed using the following three items: “In the past 7 days, I felt tired," “In the last 7 days, I had a hard time starting things because I felt tired," and “In the past 7 days, how tired do you feel on average?”. Participants answered each item on a scale from 1 (*not at all*) to 5 (*very much*). The items were averaged (Cronbach's α = 0.90).

Pain intensity was assessed using the following item: “How would you rate your average pain level?”. Participants answered on a scale from 1 (*no pain*) to 5 (*worst pain imaginable*).

Measures of additional health-related variables such as perceived mobility, limitations in activities of daily living, limitations in instrumental activities of daily living, global physical and mental health, life

satisfaction, indication for enrollment in the outpatient CR, left ventricular ejection fraction (LVEF), maximal aerobic power (MAP), abdominal circumference, glycated hemoglobin (HbA1c), triglycerides, low-density lipoprotein (LDL)-cholesterol, blood pressure, smoking, and the presence of comorbidity are described in Supplementary Material 3.

Statistical Analyses

Descriptive Statistical Analyses

First, we conducted descriptive analyses of the participants' characteristics and compared them between the outpatient and control participants using independent *t* tests for numerical variables and chi-squared tests for categorical variables. Additional analyses were carried out to compare outpatients' motivational and health-related characteristics measured at the beginning and at the end of the CR using independent *t* tests. Third, we estimated the levels of PA and SB for seven consecutive days during the week following discharge from the CR program for the outpatients and for seven consecutive days for control participants. Fourth, we compared the levels of PA and SB in outpatients with those in control participants using independent *t* tests.

Univariate and Multivariate Regression Analyses

First, we conducted univariate and multivariate regression analyses to examine the associations of motivational (i.e., perceived capabilities to engage in PA, intention to engage in PA, instrumental and affective attitudes toward PA, and approach-avoidance tendencies) and emotional health-related factors (i.e., depressive symptoms, anxiety, fatigue, and pain intensity) with PA and SB levels. Outpatients' motivational and emotional health-related factors measured at the end of the CR were used in the univariate and multivariate analyses. In the univariate and multivariate regression analyses, interaction terms were tested between participant type (i.e., outpatient vs. control) and the motivational and emotional health-related factors to examine whether the associations of motivational and emotional health-related factors with the levels of PA and sedentary behavior SB differed between outpatients and control participants. The independent variables were standardized. Finally, logistic regression analyses were conducted to determine the odds ratio associated with meeting the recommended level of PA (i.e., 150 min of MVPA per week [10]) based on the motivational and emotional health-related factors. Statistical assumptions, including normality of residuals, linearity, multicollinearity, and undue influence, were checked, and met for all models using the Performance R package [35].

Results

Descriptive results

Table 1 shows the results of the sample characteristics. A total of 119 participants (68 patients enrolled in an CR program [86.76% males; M_{age} = 57.76 ± 10.76 years] and 51 community-dwelling control participants [45.10% males; M_{age} = 57.35 ± 6.33 years]) were included in the study. The mean patient

LVEF was 52.87% (± 10.49), and most patients were enrolled in the CR program for acute coronary syndrome (40.30% non-ST elevation and 26.87% ST elevation). Before the cardiovascular event that prompted CR, outpatients self-reported significantly lower PA levels than control participants. The emotional health-related variables showed that at the beginning of the CR, outpatients showed significantly higher perceived fatigue. Outpatients self-reported a significantly higher number of IADLs, greater difficulty with mobility, lower global physical and mental health, and life satisfaction compared to control participants. The PA-related motivational variables showed that at the beginning of the CR, outpatients had significantly lower perceived capability and affective attitudes than control participants. No statistically significant differences were observed in approach-avoidance tendencies toward PA and SB between outpatients and control participants.

Table 1
Descriptive Characteristics of the Outpatients at the Beginning of the Cardiac Rehabilitation Program and Before the Physical Activity Measures for the Community-Dwelling Participants

Characteristics **N = 119**	**Outpatients** **N = 68** **(42.2%)**	**Controls** **N = 51** **(31.7%)**	
	M (SD)	M (SD)	*P* value*
Demographics and anthropometrics			
Sex (*n*, %)			
Female	9 (13.24%)	28 (54.90%)	
Male	59 (86.74%)	23 (45.10%)	< .001
Age (years, *n*, %)			
< 50	14 (20.90%)	5 (9.80%)	
50–59	25 (37.31%)	31 (60.78%)	
60–69	17 (25.37%)	12 (23.53%)	
70–79	10 (14.93%)	3 (5.88%)	
> 80	1 (1.49%)	0 (0%)	.074
Body mass index (Kg/m^2; *n*, %)			
Underweight < 18.5	0 (0%)	2 (3.92%)	
Normal 18.5–<25	19 (27.79%)	25 (49.01%)	
Overweight 25–<30	30 (45.45%)	19 (37.25%)	
Obese ≥ 30	17 (25.76%)	5 (9.80%)	.018
Usual level of PA (*n*, %)			
Physically inactive	18 (26.47%)	3 (6.38%)	
Some PA	28 (41.18%)	18 (38.30%)	
Regular PA	20 (29.41%)	23 (48.94%)	
High training	2 (2.94%)	3 (6.38%)	.020
Emotional health-related variables			
Anxiety [1–5]	1.79 (0.90)	1.60 (0.87)	.243
Depressive symptoms [1–5]	1.62 (0.76)	1.58 (0.74)	.722

Characteristics N = 119	Outpatients N = 68 (42.2%)	Controls N = 51 (31.7%)	
Fatigue [1–5]	2.55 (1.06)	2.01 (0.85)	.003
Pain intensity [1–5]	2.10 (2.29)	1.62 (1.89)	.226
Additional health-related variables			
Number of ADL (*n*, %)			
No ADL	59 (86.76%)	44 (97.78%)	
> 1 ADL	9 (13.24%)	1 (2.22%)	.094
Number of IADL (*n*, %)			
No IADL	49 (72.13%)	41 (83.67%)	
> 1 IADL	18 (26.87%)	8 (16.32%)	.026
Mobility	3.70 (0.54)	3.95 (0.23)	.002
Global physical health [1–5]	2.88 (0.89)	3.69 (0.82)	< .001
Global mental health [1–5]	3.77 (0.98)	4.10 (0.76)	.046
Life satisfaction [1–7]	5.68 (1.26)	6.24 (0.66)	.005
			(continued)

Table 1
(*continued*)

Characteristics **N = 119**	**Outpatients** **N = 68** **(42.2%)**	**Controls** **N = 51** **(31.7%)**	
	M (SD)	M (SD)	*P* value*
LVEF (%)	52.87 (10.49)	Na.	
MAP (Watt)	141.08 (45.08)	Na.	
Abdominal circumference (cm)	99.31 (12.02)	Na.	
HbA1c (%)	5.69 (0.74)	Na.	
Triglycerides (mmol/L)	1.59 (1.08)	Na.	
LDL (mmol/L)	2.94 (1.25)	Na.	
Systolic blood pressure (mmHg)	126.10 (14.64)	Na.	
Diastolic blood pressure (mmHg)	72.49 (8.29)	Na.	
Cigarette (pack/year)	10.82 (14.85)	Na.	
Comorbidity (*N* = 67, *n*, %)		Na.	
0	9 (13.43%)	Na.	
1	18 (26.87%)	Na.	
≥ 2	40 (59.70%)	Na.	
Indication for enrollment in the outpatient CR (*N* = 67, *n*; %)			
Acute coronary syndrome NSTEMI	27 (40.30)	Na.	
Acute coronary syndrome STEMI	18 (26.87)	Na.	
Acute coronary syndrome UA	4 (5.97)	Na.	
Coronary artery bypass graft surgery	7 (10.45)	Na.	

Abbreviations: M, mean; SD, standard deviation; PA, physical activity; ADL, limitations in activities of daily living; IADL, limitations in instrumental activities of daily living; CR; cardiac rehabilitation; NSTEMI, non-ST elevation myocardial infarction; STEMI, ST elevation myocardial infarction; UA, unstable angina; ACS, acute coronary syndrome; LVEF, left ventricular ejection fraction; MAP, maximal aerobic power; HbA1c, fasting glycated hemoglobin; LDL, low-density lipoprotein-cholesterol; SB, sedentary behaviors. Usual level of PA was assessed with the Saltin-Grimby PA Level Scale. A positive score in *Approach toward PA* suggests a tendency to approach PA, while a negative score suggests a tendency to avoid PA. A positive score in *Approach toward SB* suggests a tendency to approach PA, while a negative score suggests a tendency to avoid PA. **P*-value of the difference between the outpatients and control participants.

Characteristics N = 119	Outpatients N = 68 (42.2%)	Controls N = 51 (31.7%)	
Coronary artery disease without ACS or surgery	8 (11.94)	Na.	
Heart failure	2 (2.99)	Na.	
Valvular surgery	1 (1.49)	Na.	
Motivational variables			
Perceived capabilities [1–5]	4.21 (0.97)	4.90 (0.36)	< .001
Intention [1–10]	5.51 (1.34)	4.99 (1.50)	.053
Instrumental attitudes [1–10]	9.09 (1.42)	9.36 (1.55)	.325
Affective attitudes [1–10]	8.16 (2.06)	8.96 (1.60)	.024
Approach toward PA	-34.22 (109.97)	-20.82 (63.82)	.461
Approach toward SB	8.98 (102.38)	-20.59 (77.78)	.111

Abbreviations: M, mean; SD, standard deviation; PA, physical activity; ADL, limitations in activities of daily living; IADL, limitations in instrumental activities of daily living; CR; cardiac rehabilitation; NSTEMI, non-ST elevation myocardial infarction; STEMI, ST elevation myocardial infarction; UA, unstable angina; ACS, acute coronary syndrome; LVEF, left ventricular ejection fraction; MAP, maximal aerobic power; HbA1c, fasting glycated hemoglobin; LDL, low-density lipoprotein-cholesterol; SB, sedentary behaviors. Usual level of PA was assessed with the Saltin-Grimby PA Level Scale. A positive score in *Approach toward PA* suggests a tendency to approach PA, while a negative score suggests a tendency to avoid PA. A positive score in *Approach toward SB* suggests a tendency to approach PA, while a negative score suggests a tendency to avoid PA. **P*-value of the difference between the outpatients and control participants.

Physical Activity and Sedentary Behavior Levels One Week After the Cardiac Rehabilitation

Table 2 presents the PA and SB levels for the outpatients and control participants. During the first week following the rehabilitation period, only 17 out of 68 outpatients and 42 out of 51 control participants had a valid accelerometer data (see Fig. 1 for the flow diagram). As a result, the PA and SB levels were analyzed for 59 participants. The results showed that, in the week following the CR, the outpatients spent an average of 548.69 (± 58.64) min per day in SB, 211.16 (± 60.83) min in light-intensity PA, 57.85 (± 34.68) min in moderate-intensity PA, 2.28 (± 3.67) min in vigorous-intensity PA. Thus, on average, outpatients spent 60.21 (± 34.79) min per day in MVPA, with 88.2% of them meeting the recommended weekly level of 150 min of MVPA. Control participants displayed a similar overall pattern, although they spent slightly more time in light-intensity PA (253.77 ± 69.11, $p = .035$) and slightly less time in moderate-intensity PA (37.46 ± 25.03, $p = .014$). On average, control participants spent significantly less time (41.78 ± 2.088, $p = .038$) per day in MVPA compared to outpatients, with 71.4% of them meeting the

recommended 150 min per week of MVPA (p = .310). Finally, outpatients averaged 10083.00 (± 4493.27) steps per day, while the controls averaged 8785.02 (± 3424.07) steps per day. However, this difference was not statistically significant (p = .237).

Table 2
Averaged Daily Physical Activity and Sedentary Behavior Levels During the First Week After the Cardiac Rehabilitation Program and During a Usual Week for the Community-Dwelling Participants

Levels (min/day)	**Outpatients (N = 17*)**	**Controls (N = 42*)**	
	M (SD)	M (SD)	*P* value**
Sedentary behavior	548.69 (58.64)	552.26 (78.25)	.866
Light activity	211.16 (60.83)	253.77 (69.11)	.035
Moderate activity	57.85 (34.68)	37.46 (25.03)	.014
Vigorous activity	2.28 (3.67)	3.66 (7.80)	.490
MVPA	60.21 (34.79)	41.78 (28.08)	.038
Number of steps	10083.00 (4493.27)	8785.02 (3424.07)	.237
Meeting the weekly recommended level (150 min MVPA); *n* (%)	15 (88.24%)	30 (71.43%)	.310

Abbreviations: M, mean; SD, standard deviation; MVPA, moderate-to-vigorous physical activity. *The sample size was reduced because some participants did not complete all the measures. ***P* value of the difference between the different participant types (outpatients vs. controls). Recommendation is of at least 150 min of MVPA per week [10].

Relationships of Motivational and Emotional Health-Related Variables with Physical Activity and Sedentary Behavior Levels

Accelerometer-based MVPA and sitting time were used in the primary analyses, while accelerometer-based steps per day, self-reported MVPA, and self-reported time spent sitting were included in the secondary analyses and can be found in Supplementary Materials 4 and 5 (Tables S1, S2, and S3). Table 4 presents the results from both univariate and multivariate regression analyses.

Univariate analyses. The results of univariate analyses (Table 3) revealed that outpatients had higher levels of MVPA (b = 18.43, 95% CI [1.10; 35.77], R^2 adjusted = 0.06, p = .038), and that affective attitudes toward PA positively predicted daily time spent in MVPA (b = 10.32, 95% CI [1.09; 19.55], R^2 adjusted = 0.07, p = .029). No other significant associations with MVPA were found. Additionally, sex (ref. female; b = 47.54, 95% CI [11.41; 83.67], R^2 adjusted = 0.09, p = .011) and pain intensity (b = -20.44, 95% CI [-38.66;

-2.22], R^2 adjusted = 0.07, p = .029) predicted time spent in SB. Finally, higher perceived capability to engage in PA (OR = 1.17, 95% CI [1.02; 1.34], p = .030*)*, stronger instrumental (OR = 1.11, 95% CI [1.03; 1.19], p = .006), and affective attitudes toward PA (OR = 1.26, 95% CI [1.12; 1.41], p < .001) increased the likelihood of meeting the recommended level of 150 min of weekly PA. Notably, no significant interactions were observed between the PA or SB predictors and participant type (i.e., outpatient vs. control; *ps* > .05, see Supplementary Material 6, Table S4).

Table 3. Associations of Motivational and Emotional Health-Related Variables with Physical Activity and Sedentary Behavior Levels for One Week

Outcome: daily MVPA	Univariate models (N =59)		Multivariate model (N =59)	
	b (95% CI)	*P* value	b (95% CI)	*P* value
Intercept			40.66 (31.55; 49.77)	<.001
Participant type (ref. controls)				
Outpatients	18.43 (1.10; 35.77)	.038	18.81 (1.58; 36.05)	.047
R^2 adjusted	0.06			
Age	-0.32 (-10.76; 10.12)	.951		
Sex (ref. Female)				
Male	0.90 (-15.5; 17.2)	.917		
Body mass index	-4.47 (-14.23; 5.30)	.364		
Intention toward PA	3.94 (-3.96; 11.83)	.322		
Perceived capabilities toward PA	8.52 (-1.96, 18.99)	.109		
Instrumental attitudes	3.74 (-3.82; 11.30)	.326		
Affective attitudes	10.32 (1.09; 19.55)	.029	10.15 (1.21; 19.09)	.049
R^2 adjusted	0.07			
Approach toward PA	2.60 (-4.79; 9.99)	.484		
Approach toward SB	5.50 (-1.59, 12.60)	.126		
Depressive symptoms	1.02 (-7.77, 9.81)	.817		
Anxiety	-0.40 (-8.79; 7.99)	.9244		
Fatigue	-3.18 (-13.64; 7.29)	.546		
Pain intensity	-4.02 (-12.10; 4.06)	.323		
R^2 adjusted for the multivariate model			0.12	

(continued)

Table 3. (*continued*)

Outcome: daily SB	Univariate models (*N* = 59)		Multivariate models (*N* = 59)	
	b (95% CI)	*P* value	b (95% CI)	*P* value
Intercept			534.90 (506.75; 563.03)	<.001
Participant type (ref. control)				
Outpatients	-3.57 (-45.75; 38.61)	.866	-17.49 (-59.14; 24.17)	.404
Age	-0.38 (-24.84; 24.07)	.975		
Sex (ref. Female)				
Male	47.54 (11.41; 83.67)	.011	40.47 (1.02; 79.92)	.045
R^2 adjusted	0.09			
Body mass index	0.14 (-22.90; 23.19)	.999		
Intention toward PA	-8.88 (-27.21; 9.46)	.336		
Perceived capabilities toward PA	11.17 (-13.68; 36.02)	.372		
Instrumental attitudes	-3.05 (-20.84; 14.74)	.732		
Affective attitudes	-15.07 (-37.20; 7.06)	.178		
Approach toward PA	-4.73 (-24.47; 15.02)	.633		
Approach toward SB	-16.24 (-35.05; 2.57)	.089		
Depressive symptoms	-0.56 (-21.08; 19.97)	.957		
Anxiety	-3.85 (-23.41; 15.70)	.695		
Fatigue	-11.45 (-35.77; 12.88)	.350		
Pain intensity	-20.44 (-38.66; -2.22)	.029	-14.19 (-33.13; 4.74)	.139

R^2 adjusted	0.07	
R^2 adjusted for the multivariate model		0.10

(continued)

Table 3. (*continued*)

Outcome: meeting the recommended levels of MVPA	Univariate models (N = 59)		Multivariate models (N = 59)	
	OR (95% CI)	*P* value	OR (95% CI)	*P* value
Intercept			1.90 (1.69; 2.13)	<.001
Participant type (ref. control)				
Outpatients	1.18 (0.93; 1.50)	.175	1.23 (0.99; 1.51)	.066
Age	1.02 (0.88; 1.17)	.809		
Sex (ref. Female)				
Male	1.10 (0.88; 1.37)	.415		
Body mass index	0.91 (0.80; 1.03)	.153		
Intention toward PA	1.10 (0.99; 1.22)	.090		
Perceived capabilities toward PA	1.17 (1.02; 1.34)	.030	1.19 (1.05; 1.35)	.007
Instrumental attitudes	1.11 (1.03; 1.19)	.006	1.02 (0.90; 1.15)	.804
Affective attitudes	1.26 (1.12; 1.41)	<.001	1.24 (1.06; 1.45)	.011
Approach toward PA	0.98 (0.87; 1.10)	.678		
Approach toward SB	1.07 (0.95; 1.19)	.278		
Depressive symptoms	1.01 (0.90; 1.14)	.836		
Anxiety	1.01 (0.91; 1.13)	.817		
Fatigue	1.06 (0.92; 1.22)	.430		
Pain intensity	1.00 (0.89; 1.11)	.957		

Abbreviations: PA, Physical activity; SB, sedentary behaviors; MVPA, Moderate-to-vigorous PA; 95% CI, Confidence interval at 95%; OR, odd ratio. The adjusted model includes only the significant predictors of

the univariate models, with the exception of the participants type (i.e., patients versus control) which was adjusted for. The recommended levels of MVPA have been set at greater than or equal to 150 min of MVPA per week.[10] The R^2 adjusted was calculated only for statistically significant predictors [10]. The R^2 adjusted was calculated only for statistically significant predictors.

Multivariate models. Affective attitudes (b = 10.15, 95% CI [1.21; 19.09], p = .049) remained significantly associated with daily time spent in MVPA, after controlling for participant type. The final model explained 12% of the variance in MVPA. Sex remained associated with SB time (ref. female, b = 40.47, 95% CI [1.02; 79.92], p = .045), after controlling for pain intensity (b = -14.19, 95% CI [-33.13, 4.74], p = .139) and for participant type (b = -17.49, 95% CI [-59.14; 24.17], p = .045). The final model explained 10% of the variance in SB. Finally, perceived capabilities (OR = 1.19, 95% CI [1.05; 1.35], p = .007) and affective attitudes toward PA (OR = 1.24, 95% CI [1.06; 1.45], p = .011) remained significantly predictive of meeting the recommended 150 min of MVPA per week, after controlling for instrumental attitudes, and participant type. However, instrumental attitudes became non-significant (p = .804). Notably, no significant interactions were observed between PA or SB predictors and participant type (i.e., outpatient vs. control; *ps* > .05, see Supplementary Material 6, Table S4).

Sensitivity Power Analyses

Following Lakens' [36] recommendations, a sensitivity curve was plotted for a linear univariate regression, with an alpha level of .05 and a sample size of 59 participants (i.e., number of participants with complete device-based data). This allowed us to assess the effect sizes that can be detected across a range of desired power levels [36]. Powers between 33%, which aligns with the limit of studies that are considered insufficiently powerful [37], and 90% were analyzed. For an univariate regression, the sensitivity curve showed that our study had between 33% and 90% power of detecting an effect size between f^2 = .04; or R^2 = .04 and f^2 = .18; or R^2 = .16, respectively. For a multivariate regression with two predictors (i.e., affective attitudes and participant type), the sensitivity curve showed that our study had between 33% and 90% power of detecting an effect size between f^2 = .06; or R^2 = .06 and f^2 = .25; or R^2 = .20, respectively. For a multivariate regression with three predictors (i.e., sex, pain intensity, participant type), the sensitivity curve showed that our study had between 33% and 90% power of detecting an effect size between f^2 = .07; or R^2 = .06 and f^2 = .29; or R^2 = .23, respectively. Additionally, a sensitivity curve was plotted for a logistic regression, with an alpha level of .05 and a sample size of 59 participants. The sensitivity curve showed that, for a positive effect of the independent variable on PA and SB levels, our study had between 33% and 90% power of detecting an effect size between OR = 1.50 and OR = 2.70, respectively. The sensitivity curve showed that, for a negative effect of the independent variable on PA and SB levels, our study had between 33% and 90% power of detecting an effect size between OR = 0.66 and OR = 0.37, respectively. The sensitivity curves are available in Supplementary Material 7, Figures S1 and S2, and are further discussed in the discussion.

Discussion

The main findings of the present study were that, during the first week following discharge from CR, outpatients were significantly more physically active than control participants in a typical week. Motivational factors (i.e., perceived capability and affective attitudes toward PA) were associated with PA, and males spent more time in SB than females. However, no emotional-health related factors were significantly associated with PA and SB. This finding of higher PA levels in CR patients compared with healthy controls suggests that the rehabilitation program is effective in promoting PA levels at least in the short term—although this difference must be interpreted with caution because the groups differ on many demographic and motivational variables. However, we were unable to identify the key components of the intervention responsible for this effect.

During the first week following discharge from CR, 88.2% of outpatients achieved the recommended 150 min of MVPA per week. These results are consistent with Bierbauer et al. [8] findings, which demonstrate that about 70% of patients succeeded in meeting PA recommendations in the first three weeks following a CR program. Additionally, our findings indicate that outpatients engaged in 18 min more of MVPA per day than control participants ($p = .038$). Although Barker et al.[38] have shown that individuals with cardiovascular disease remain less active than community-dwelling individuals without chronic disease, Steca et al. [39] demonstrated that the proportion of coronary patients who achieved the recommended PA level increased from 35.6% prior to the CR program to 60% six months after discharge. Thus, during the first week of a CR program, patients tend to be more active than control participants as they are encouraged to improve their PA levels and reduce their SB during the CR program [3]. Finally, our estimation of outpatients meeting the PA guidelines after discharge from CR exceeded the literature (88.2%). This gap may also be due to a selection bias, as only 25% of the 68 included outpatients wore an accelerometer for one week at the end of the 6-week rehabilitation period.

At the beginning of the CR, outpatients had significantly lower perceived capabilities and affective attitudes toward PA and significantly higher levels of depressive symptoms, fatigue, and pain, compared to control participants. These findings are consistent with current literature that highlights the high prevalence of pain, discomfort, and depression among cardiac patients [7, 40]. Although we found significant differences in some deliberative motivational factors, such as perceived capability and affective attitudes, between outpatients and control participants, we did not find any differences in the levels of automatic approach-avoidance tendencies toward PA and SB between the two groups. These results are difficult to compare to the existing literature, as to the best of our knowledge, no study has directly investigated and compared such automatic motivational factors between patients and control participants. Yet, it has been suggested that such automatic factors could be negatively biased in patients because of fear, pain, and discomfort felt during exercise [21, 41, 42]. For instance, the fear of PA, which is particularly pronounced in cardiac patients [43], may produce automatic avoidance tendencies toward PA [44]. The inability to observe differences in our study can be explained by several factors. First, patients enrolled in the IMPACT trial (i.e., a 6-week study aiming to improve their PA behaviors [21]) may have been more inclined to engage in regular PA than patients who were not interested in the study. Specifically, although outpatients and control participants had different perceived capabilities and affective attitudes, both groups scored highly (> 4 out of 5 and > 8 out of 10,

respectively). Second, the features of the task used to capture that approach-avoidance tendencies reactions may also explain the current findings. Participants were asked to react to the format of the picture (portrait vs. landscape) rather than its content (avatar in active vs. passive behavior). This task has been found to produce smaller effect sizes compared to tasks that require a reaction to the content of the picture [45]. Consequently, while previous studies demonstrated that healthy participants exhibited a positive approach bias toward PA stimuli and a negative approach bias toward SB stimuli [46–50], our current results, as well as a recent fMRI study relying on the same task [51], failed to reproduce such observation in healthy participants.

No significant association was found between emotional health-related factors and device-measured MVPA during the first week after discharge from CR, or during a typical week for control participants. These findings differ from previous literature, indicating that patients experiencing pain, anxiety, depression, and fatigue were less physically active [7, 23–25]. For instance, Dagner et al. [7] found that patients with low anxiety and depression had higher odds of engaging in PA for more than 30 minutes per week following discharge from CR (OR = 0.6). Conversely, patients with high pain had a lower likelihood of engaging in PA (OR = 2.00). Additionally, Andersson et al. [24] found that individuals with chronic obstructive pulmonary disease with high fatigue had lower odds of engaging in PA (OR = 2.33). Our sensitivity analyses revealed that the current study had approximately 43% power to detect an odds ratio (OR) of 0.6, 65% power to detect an OR of 2.0 and 78% to detect an OR of 2.33 (Supplementary Material 9, Figure S2). Thus, this contrast could be explained in part by our low statistical power to detect the true effect.

In contrast to emotional health-related factors, several deliberative motivational factors were associated with MVPA. Specifically, higher perceived capabilities, as well as stronger instrumental and affective attitudes toward PA, increased the odds of meeting the recommended levels of 150 min of MVPA per week. These results are consistent with Bermudez et al.'s study [2], indicating that patients exhibiting more positive affective attitudes toward PA were inclined to engage in higher PA levels after CR discharge. We found no evidence of a predictive effect of automatic approach-avoidance tendencies toward PA and SB on device-measured MVPA and sedentary behaviors in both cardiac patients and healthy control participants. These findings contrast with previous work suggesting that automatic processes, including approach-avoidance tendencies, are instrumental to the regulation of PA behaviors [14–16]. For example, studies examining automatic processes have shown that PA cues attract attention [52–54], elicit positive affective responses,[20, 55] and prompt approach tendencies [46, 50, 56], particularly in highly active individuals. These results suggest that automatic responses to PA cues that are not positive enough may partially explain the lack of engagement in PA. This lack of association in the present study can again be attributed to the task's characteristics, the relatively small sample size (*N* = 59) and the lack of statistical power. Therefore, before drawing definite conclusion about the absence of association between approach-avoidance tendencies and PA and SB behaviors, additional well-powered studies using different reaction-time tasks are needed.

Regarding device-based measured SB, male participants spent significantly more time in SB than female participants. Additionally, we observed a significant negative association between pain intensity and SB, indicating that higher pain intensity was linked to lower levels of SB. However, this association lost its significance after adjusting the model for participant type and sex (p = .139). This finding contrasts with O'Leary et al.'s study [57], which found a positive correlation between pain intensity and SB in rheumatoid arthritis patients (r = 0.31). Similar to the present study, this association lost its significance when other factors, such as age and gender, were included in multivariate models. Overall, these results suggest that the influence of PA-related motivational factors and emotional health-related factors is limited in explaining SB levels. However, it is important to note that most of the factors measured in this study related to PA and not to SB (e.g., perceived capabilities toward PA). Although the other studies have also used PA-based motivational measures to predict SB [58, 59], this feature may explain the lack of significant association. Future studies should investigate a wider range of factors that directly target SB, in order to better understand the influence of motivational and emotional health-related factors on SB [58]. Finally, no significant interactions were observed between the motivational and emotional health-related predictors with levels of PA or SB and participant type (i.e., outpatient vs. control). Although this finding should be interpreted with caution given the rather small sample size of patients (n = 17), it could suggest that the motivational mechanisms underlying the regulation of PA behaviors may be identical between CR patients and community-dwelling individuals.

Constraints on Generality and Strengths

This study has several limitations that limit the generalizability of the findings. First, it used a small sample size to measure PA and SB. The sample consisted of 17 CR patients and 42 community-dwelling participants, who had significantly different baseline characteristics. A larger sample of patients would provide more accurate estimations of PA and SB in this specific population. Second, all patients involved in this study were enrolled in programs that took place at the Geneva University Hospitals, limiting the generalizability of the results to patients in other CR settings. Third, as it is common in longitudinal studies, we cannot dismiss the possibility of selective attrition bias. This means that patients who are more motivated to make lifestyle changes may be more likely to continue participating in this study compared to those who are less motivated. This bias may, for instance, account for the relatively high proportion of CR patients who were able to achieve the weekly recommended level of 150 min of MVPA. Consequently, future studies capable of retaining a larger proportion of patients may generate findings that are more representative of the general population. Fourth, while we used device-based measures of SB with the Actigraph GT3X, int is worth noting that the activPal™ device has been identified as the most reliable device for measuring SB levels [60, 61]. Fifth, despite observing significant associations between motivational factors, particularly affective attitudes, and PA behaviors, a large amount of variance remains to be explained. Therefore, the absence of predictive validity of the other motivational and emotional health-related factors needs to be interpreted with caution, as underpowered analyses may distort the size of the estimated effects [62, 63].

Nonetheless, the study has several strengths that offset these limitations. First, the study incorporated a sample of CR patients along with a group of community-dwelling participants. This characteristic allowed for the observation of PA and SB in various contexts, including leisure time after discharge from CR and the daily lives of individuals without specific medical care. Second, this study focused not only on PA but also on SB. While it is believed that both PA and SB have distinct impacts on the potential health benefits of rehabilitation [11–13], few studies have explored these behaviors simultaneously. Third, this study utilized device-based measures for PA and SB, which render more reliable estimates than self-reported questionnaires [64–66]. Fourth, this study investigated the potential motivational factors that regulate PA and SB at both the deliberative (i.e., intention, perceived capabilities, and attitudes) and automatic (i.e., approach-avoidance tendencies) levels. Such research has been scarce in the existing literature and may offer a more comprehensive perspective on the determinants of these behaviors.

Conclusion

The present study found that patients who had completed the outpatients CR program demonstrated higher levels of PA in their daily lives during the week following their discharge, compared to community-dwelling individuals. Additionally, perceived capabilities and instrumental and affective attitudes were significantly and positively associated with MVPA across all participants. These findings support the implication of motivational factors in PA behaviors among both patients and community-dwelling individuals. Future interventions should be encouraged to target these factors to promote PA levels following CR programs. Although the finding suggests that the motivational mechanisms underlying the regulation of PA may be identical between patients and individuals without medical care, future studies with larger sample sizes are needed to provide more convincing evidence regarding this possibility. Finally, we encourage future studies investigating SB correlates among patients to include SB-specific motivational factors (e.g., "*During the next two weeks, I intend to spend no more than four hours in SB a day during my leisure-time.*") and employ precise SB assessment tools such as activPal™.

Abbreviations

CR	Cardiac rehabilitation
HbA1c	glycated hemoglobin
IMPACT	Improving Physical Activity trial
LDL	low-density lipoprotein
LVEF	Left ventricular ejection fraction
MAP	Maximal aerobic power
MVPA	Moderate-to-vigorous physical activity

PA	Physical activity
PROMIS	Patients-Reported Outcomes Measurement Information System
SB	Sedentary behavior
VAAST	Visual-Approach/Avoidance-by-the-Self-Task

Declarations

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Author Contribution

The first draft of the manuscript was written by L.F. and B.C. and all authors commented on the previous versions of the manuscript. All authors read and approved the final manuscript. L.F. contributed to conceptualization, methodology, formal analysis, investigation, writing—original draft; E.T. contributed to writing–review and editing, resources; C.C. contributed to writing–review and editing, investigation; P.S. contributed to supervision, writing–review and editing; P.M. contributed to writing–review and editing, resources; C.L. contributed to writing–review and editing, resources; E.H.D contributed to writing–review and editing, resources; B.C. contributed to conceptualization, methodology, supervision, formal analysis, investigation, writing—original draft.

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Availability of Data and Materials

Deidentified data, data management, analysis codes and research materials have been made publicly available on Zenodo (https://doi.org/10.5281/zenodo.11072811).

Ethics Approval and Consent to Participate

The present study was conducted in accordance with the Declaration of Helsinki, and approved by the Ethics Committee of Geneva Canton, Switzerland (reference number: CCER2019-02257). Prior to participation in the study, all participants signed an informed consent form.

Consent for Publication

Consent for Publication declaration: not applicable.

Competing Interests

The authors have no conflicts of interest relevant to this article to disclose.

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Figures

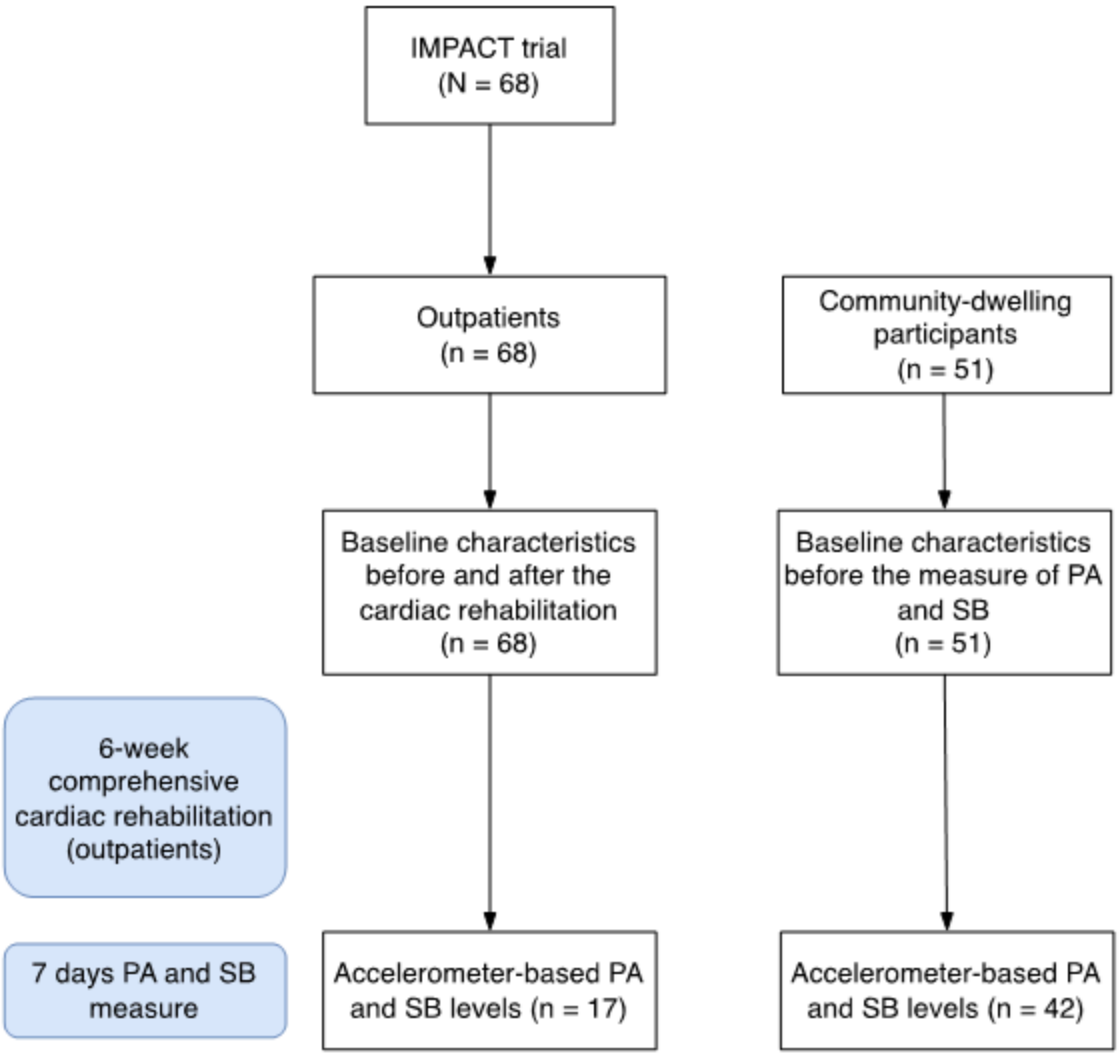


Figure 1

Flow Diagram of the Study Protocol

Abbreviations: PA, physical activity; SB, sedentary behavior. The data for outpatients comes from the Improving Physical Activity (IMPACT) trial [21]. The baseline characteristics included demographics and anthropometrics information, usual level of PA, number of limitations in activities of daily living (ADL), of limitations in instrumental activities of daily living IADL), mobility, motivational and health-related variables, and health-related information.

Supplementary Files

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